

Probing the Degeneracy of Evolving Dark Energy and Extended Neutrino Cold Dark Matter Models using DESI DR2

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Lambda cold dark matter (Λ CDM) is the standard cosmological model that describes the evolution of our universe. Despite its success in explaining different cosmological phenomena, it has been continuously challenged by various observations, among which the deviation of its best-fit value of the Hubble constant from the local measurements is a main focus of current research. In 2025, Dark Energy Spectroscopic Instrument (DESI) claimed w_0w_a CDM model, which assumes evolving dark energy, was strongly preferred over Λ CDM according to their latest baryon acoustic oscillation (BAO) data and cosmic microwave background (CMB) data while previous research has demonstrated that Λ CDM ξ , a model that relaxes constraints on neutrino mass and chemical potential, also shows promise in addressing the Hubble tension. This project aims to probe the degeneracy in the parameters of the two models w_0/w_a and $M_\nu/\xi_{\nu 3}$, indicating that BAO data cannot distinguish between the two. The success of w_0w_a CDM therefore equally implies the success of Λ CDM ξ . A cosmological model that includes both dark energy and neutrino parameters was developed using **CLASS** theory code and Markov Chain Monte Carlo (MCMC) was employed to perform data fitting based on DESI DR2. I studied the individual dependence between each pair of parameters out of the four $w_0, w_a, M_\nu, \xi_{\nu 3}$. A perturbative approach was introduced and applied to derive the relation between each two parameters semi-analytically. The result shows a strong correlation between each pair of parameters and the correlation plots from MCMC fitting align well with the semi-analytical formula. It demonstrates the degeneracy between the two models and suggests a novel method to understand the dependence of each pair of parameters in a more analytical way rather than only relying on MCMC.

I. INTRODUCTION

Λ CDM is one of the most successful cosmological models in describing the cosmic microwave background anisotropy and other cosmological phenomena [1–3]. However, the Hubble tension, which refers to the discrepancy between the local measurements of the Hubble constant H_0 and the predicted value of the Λ CDM model [4, 5], and many recent observations continuously challenge it. One of the most promising alternatives w_0w_a CDM was found to be preferred over Λ CDM at 3.1σ when combining DESI BAO and CMB data, drawing considerable attention from the community [6]. This model features an evolving dark energy instead of a cosmological constant. Another less extensively studied model Λ CDM ξ , assuming non-zero neutrino mass and degeneracy factors $\xi_{\nu i}$, also shows promise in addressing the Hubble tension [7, 8]. Both extended models were preferred over the standard model despite the introduction of two fitting parameters w_0, w_a and $M_\nu, \xi_{\nu 3}$.

The success of both may hint at their degeneracy in the data, thereby motivating me to study their possible relation both numerically and semi-analytically. A cosmological model including both evolving dark energy and relaxed neutrino constraints was developed using **CLASS** theory code [9] and was fitted into DESI DR2 BAO data using MCMC. The correlation of w_0/w_a and $M_\nu/\xi_{\nu 3}$ was examined one by one by fixing the other two and the six standard Planck’s fitting parameters. A

perturbative approach was introduced to reproduce a similar curve that aligns with the correlation contour of the two parameters.

II. THEORY

A. Introduction to w_0w_a CDM Model

w_0w_a CDM model assumes that dark energy follows the Chevalier-Polarski-Linder (CPL) parametrization equation of state [10, 11] instead of the standard cosmological constant equation of state. The energy density is as follows

$$\rho_\Lambda(a) = \rho_{\Lambda 0} a^{-3(1+w_0+w_a)} e^{3w_a(a-1)} \quad (1)$$

where $\rho_{\Lambda 0}$ is the present-day dark energy density. This model features an evolving dark energy density in stark contrast of the traditional cosmological constant, which is recovered when $w_0 = -1$ and $w_a = 0$.

B. Introduction to Λ CDM ξ Model

In many cosmological models, $\sum m_\nu = 0.06$ eV was implicitly assumed [3, 6]. Although both Planck and DESI tried to relax the constraint on neutrino mass, they never did it together with a non-zero degeneracy factor $\xi_i = \mu_i/T_\nu$, where μ_i and T_ν are chemical potential and neutrino decoupling temperature [3, 6]. Λ CDM ξ model on the other hand allows total neutrino mass M_ν and the third degeneracy factor

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of mass eigenstate ξ_{ν_3} to vary [7, 8].

The total neutrino and anti-neutrino energy density of the three mass eigenstates are given by

$$\rho_\nu(a) = \frac{1}{2\pi^2} \sum_i \int_0^\infty \left(\frac{\sqrt{p^2 + (\frac{M_\nu}{3})^2}}{e^{\frac{pa}{T_\nu} - \xi_{\nu_i}} + 1} + \frac{\sqrt{p^2 + (\frac{M_\nu}{3})^2}}{e^{\frac{pa}{T_\nu} + \xi_{\nu_i}} + 1} \right) p^2 dp \quad (2)$$

where p is the momentum. Since $M_\nu \ll T_\nu$ is expected based on neutrino experiment, the mass contribution is ignored in the exponential function. Besides, given that data is insensitive to individual neutrino mass, M_ν is assumed to be equally shared among the three eigenstates as adopted in [3]. Assuming $\xi_{\nu_\mu} = \xi_{\nu_\tau}$ and $\xi_{\nu_e} = 0.044$ as obtained in Ref. [12], and using neutrino mixing angle values from Ref. [13], ξ_{ν_1} and ξ_{ν_2} can be expressed as a function of ξ_{ν_3}

$$\pi^2 \xi_{\nu_1} + \xi_{\nu_1}^3 = -0.219827 (\pi^2 \xi_{\nu_3} + \xi_{\nu_3}^3) + 0.640798, \quad (3)$$

$$\pi^2 \xi_{\nu_2} + \xi_{\nu_2}^3 = 0.423289 (\pi^2 \xi_{\nu_3} + \xi_{\nu_3}^3) \quad (4)$$

III. METHOD

A. Developing a Combined Model

To study the degeneracy between neutrino and dark energy parameters, both evolving dark energy and finite neutrino mass and degeneracy factor were introduced to the model. **CLASS** theory code was employed with activation of the CPL parametrization and three non-cold dark matter (ncdm) components, which act as massive neutrino species. Both of them altered the standard Friedmann equations and Boltzmann equations automatically through **CLASS** setup. The expansion history of the universe is now governed by

$$\frac{H(a)}{H_0} = \sqrt{\Omega_m a^{-3} + \Omega_\gamma a^{-4} + \Omega_\Lambda a^{-3(1+w_a)} e^{3w_a(a-1)} + \frac{\rho_\nu(a; M_\nu, \xi_{\nu_3})}{\rho_{crit}}} \quad (5)$$

B. Data Fitting

The model contains 10 parameters (6 standard Planck's parameters + 4 new parameters). Such a high-dimensional parameter space poses challenges for constraining them using MCMC and also obscures intrinsic correlation between each pair of parameters after marginalization. To disclose their dependence, I first fixed the 6 standard Planck's parameters to the Planck's best fit value and then studied the correlation between w_0/w_a and M_ν/ξ_{ν_3} one by one by fixing the other two. For example, to probe the degeneracy of w_0 and ξ_{ν_3} , I fixed w_a and M_ν . The same procedure was performed for other pairs. There is a concern that w_0/w_a and M_ν/ξ_{ν_3} may affect the 6 standard Planck's parameters differently, which is hidden when the 6 parameters are fixed. This warrants further

study.

Following the above procedure, **Cobaya** [14, 15] was employed to perform MCMC based on DESI DR2 [6]. The BAO measurements of DESI contain three observables $\beta_\perp$, $\beta_\parallel$ and β_{avg} corresponding to the inverse of angular separation, inverse of redshift separation and the angle-averaged quantity:

$$\beta_\perp(z) = \int_0^z \frac{cdz'}{r_d H(z')} \quad (6)$$

$$\beta_\parallel(z) = \frac{c}{r_d H(z)} \quad (7)$$

$$\beta_{avg}(z) = (z(\beta_\perp(z))^2 \beta_\parallel(z))^{\frac{1}{3}} \quad (8)$$

where r_d is the comoving distance of baryon sound horizon. The main focus is the correlation plot of each pair of parameters instead of their best-fit values.

C. BAO Angle Perturbation

Their correlation can also be studied semi-analytically as adopted in Ref. [7]. Assuming the effect of varying w_0/w_a can be compensated by M_ν/ξ_{ν_3} , we have

$$\delta\beta_{avg} = 0 \quad (9)$$

δ here refers to a slight change in the value of β_{avg} due to the deviation of w_0, w_a, M_ν and ξ_{ν_3} from their reference values. The multivariable Taylor expansion was then applied to find the change up to the first order of w_0, w_a, M_ν and second order of ξ_{ν_3} . See **Appendix A** for the detailed calculation. The result is in the form of

$$0 = A\delta w_0 + B\delta w_a + C \frac{\delta M_\nu}{M_\nu^{ref}} + D\delta\xi_{\nu_3} + E\delta\xi_{\nu_3}^2, \quad (10)$$

where A, B, C, D, E are the dimensionless Taylor expansion coefficients, which depend on the reference point in the 10-parameter space. In principle, since β_{avg} depends on the redshift z , so do these coefficients. Hence, I made use of the weighted average coefficients of different z based on the data uncertainty at each redshift.

IV. RESULT

The four pairs of parameters in Fig. 1 show a clear correlation. The two ξ_{ν_3} contour plots show a noticeable quadratic behavior (that is why I expanded its coefficient up to second order) while the two M_ν plots show a linear relation with w_a and w_0 . This implies the degeneracy in the two models as both M_ν and ξ_{ν_3} can mimic the effect of either w_0 or w_a at least when all other parameters are fixed. The semi-analytical formulas derived from BAO angle perturbation align well with the contour. Slight deviation is expected as it is after all

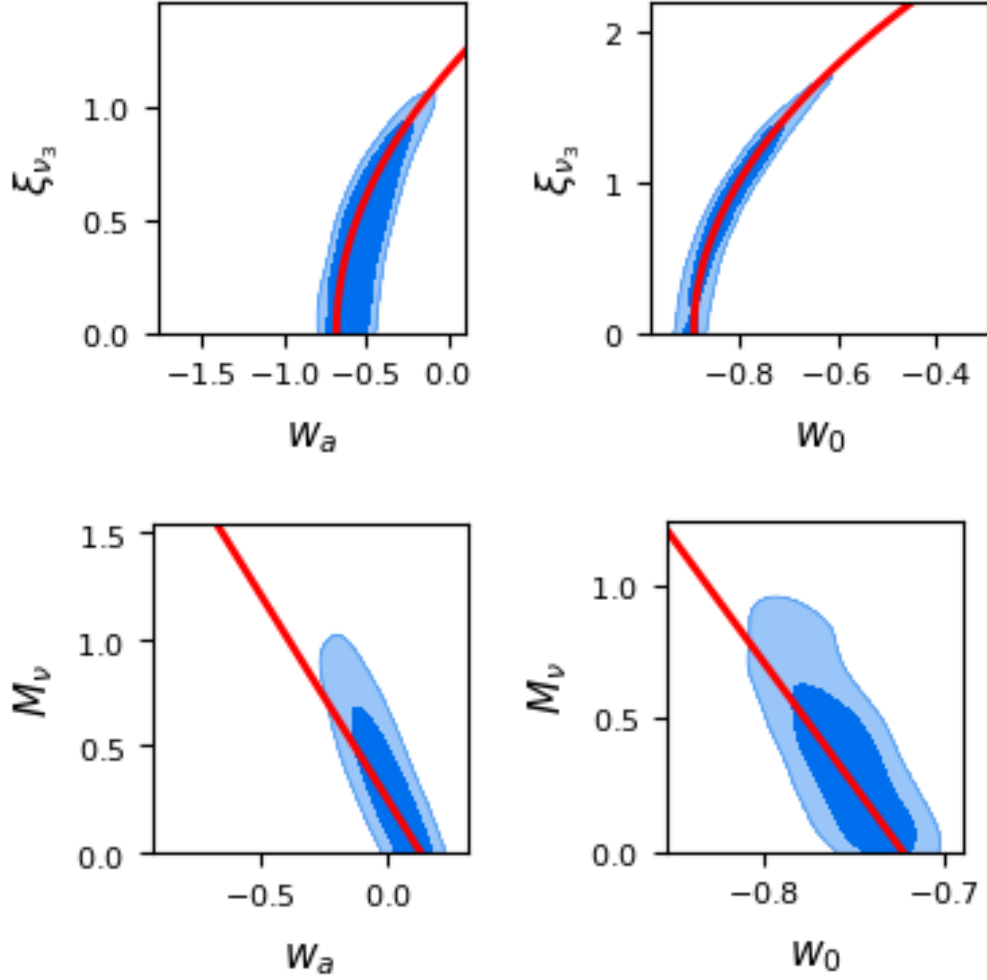


Figure 1. Correlation contours of $\xi_{\nu 3}$, w_a (upper left), $\xi_{\nu 3}$, w_0 (upper right), M_ν , w_a (bottom left), M_ν , w_0 (bottom right). The red curves represent the semi-analytical formula obtained by perturbing the BAO angle.

a Taylor expansion approximation and also only accounts for the evolution history of the universe but not the cosmological perturbation.

We can utilize the perturbation formula to understand how varying $\xi_{\nu 3}$ and M_ν can compensate the effect of w_a and w_0 . Among these four pairs, $\xi_{\nu 3}$ and w_a are of the utmost interest because $\xi_{\nu 3}$ is generally treated as zero while w_a carries an exponential effect in Eq. 1. In this preliminary study, I obtained the equivalent $\xi_{\nu 3}$ value if $w_a = 0$ i.e. if there is no significantly evolving dark energy. Assuming w_0 and M_ν are fixed to their reference point, from the semi-analytical formula, we have

$$24.99958(w_a - w_a^{ref}) = -0.29921(\xi_{\nu 3} - \xi_{\nu 3}^{ref}) + 12.88102(\xi_{\nu 3} - \xi_{\nu 3}^{ref})^2 \quad (11)$$

The equivalent value is $\xi_{\nu 3} = 1.160476$. Previous study of pure Λ CDM ξ gives a similar best-fit value $\xi_{\nu 3} \sim 1$ [8], supporting the success of the semi-analytical formula.

V. DISCUSSION

It should be noted that by obtaining this equivalent $\xi_{\nu 3}$, I have assumed specific values of w_0 and M_ν . A more systematic way, for example by generating multiple curves of different w_0 and M_ν and averaging out their equivalent $\xi_{\nu 3}$, should be explored in the future.

Another concern regarding the reference point was also raised. Currently, the choice of reference point is somewhat arbitrary. Depending on future purposes, the reference point should be carefully chosen. For example, if we aim to reproduce the same result of $w_0 w_a$ CDM using Λ CDM ξ , a better choice of reference point would be those best fit of $w_0 w_a$ CDM.

As mentioned above, this degeneracy study is limited to only the 4-parameter space. I did not account for the fact that w_0 , w_a and M_ν , $\xi_{\nu 3}$ affect the 6 standard Planck parameters

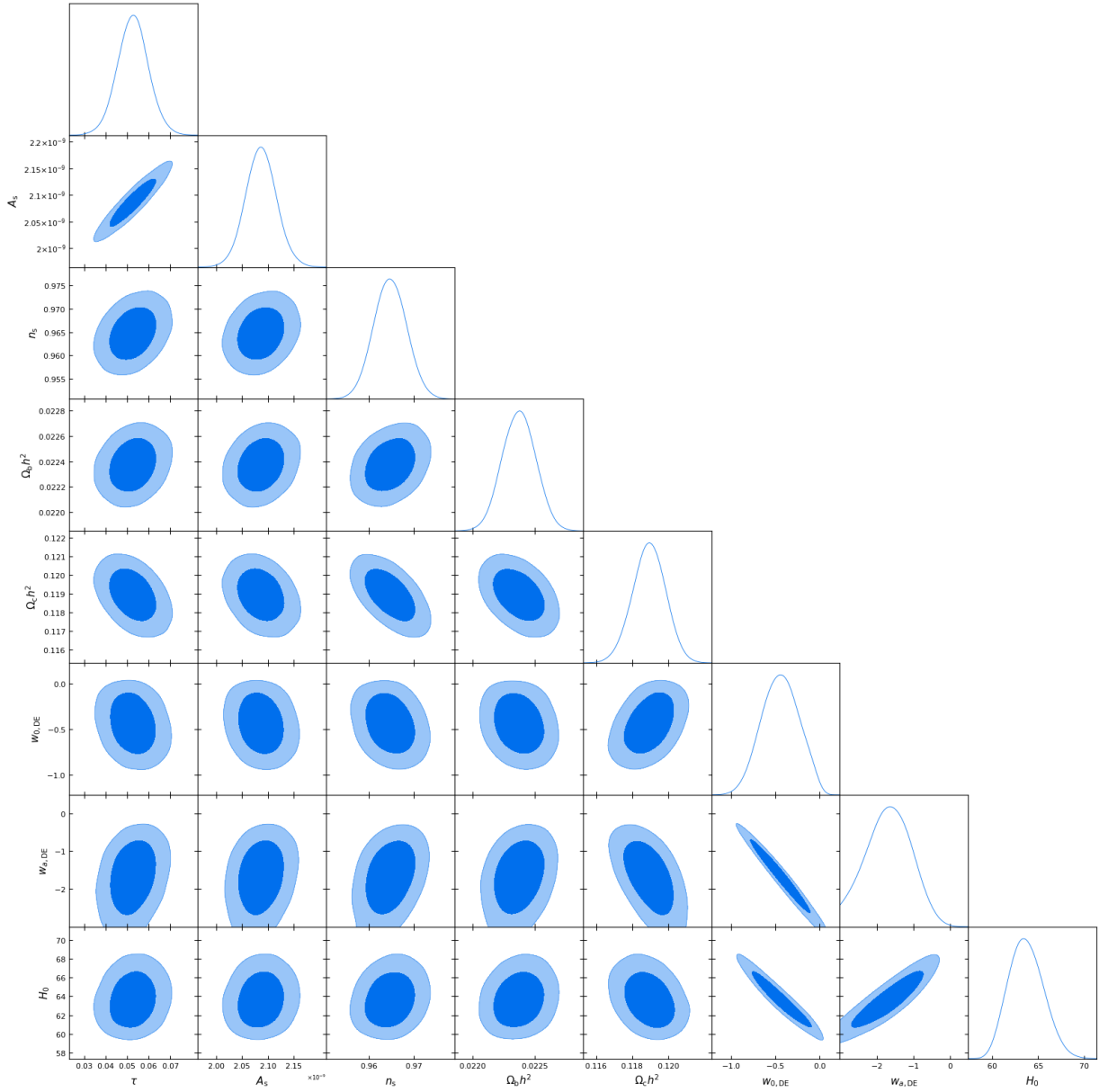


Figure 2. Triangular plot of the $w_0 w_a$ CDM model based on Planck CMB and DESI BAO data. The fitting was performed by Dr. Zhang.

differently. In fig.2, we see a clear correlation between w_0/w_a and H_0 while Yeung showed that M_ν and $\xi_{\nu 3}$ have correlation not only with H_0 , but also with other parameters depending on the data [8]. Future studies should consider the differences.

Last but not least, I focused only on BAO data in this study. As we have seen, BAO data is dominated by the effect of the evolution history of the universe. Future studies including CMB temperature and polarization data may give a more comprehensive understanding of the degeneracy including the perturbation effect.

VI. CONCLUSION

To conclude, by fixing the 6 standard Planck's parameters, I studied the degeneracy between w_0/w_a and $M_\nu/\xi_{\nu 3}$. The result shows a significant correlation between each pair of the above four parameters, showing promise in the degeneracy between $w_0 w_a$ CDM and Λ CDM ξ models. The correlation contour aligns well with the semi-analytical formula from BAO angle perturbation, proving the dominance of the evolution effect of the universe in BAO data. The formula also allows the recovery of a similar result of $\xi_{\nu 3}$ obtained by pure Λ CDM ξ fitting. It warrants future study on their independent effect on the 6 standard parameters and with CMB data.

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Appendix A: Semi-Analytical Formula from Perturbative Approach

1. Three BAO Angle and One CMB Angle Definition

$$\beta_{\perp}(z) = \int_0^z \frac{dz'}{r_d H(z')}, \quad (\text{A1})$$

$$\beta_{\parallel}(z) = \frac{1}{r_d H(z)}, \quad (\text{A2})$$

$$\beta_{\text{avg}}(z) = (z\beta_{\perp}^2(z)\beta_{\parallel}(z))^{1/3}, \quad (\text{A3})$$

$$(\text{A4})$$

2. Cosmological Equations

$$H(z) = H_0 \sqrt{\Omega_b(1+z)^3 + \Omega_c(1+z)^3 + \Omega_{\gamma}(1+z)^4 + \frac{\rho_r(z)}{\rho_{\text{crit}}} + \Omega_{\Lambda}(1+z)^{3(1+w_0+w_a)} e^{-3w_a(1-\frac{1}{1+z})}}, \quad (\text{A5})$$

$$r_d = \int_{z_d}^{\infty} \frac{c_s(z') dz'}{H(z')}, \quad (\text{A6})$$

$$\chi(z) = \int_0^z \frac{dz'}{H(z')}, \quad (\text{A7})$$

$$c_s(z) = \frac{1}{\sqrt{3(1 + \frac{3\Omega_b}{4\Omega_{\gamma}(1+z)})}} \quad (\text{A8})$$

3. Neutrino Equations

$$\rho_r(z) = \sum_i \frac{1}{2\pi^2} \int_0^{\infty} \sqrt{p^2 + \left(\frac{M_{\nu}}{3}\right)^2} p^2 \left(\frac{1}{e^{p/(T_{\nu}(1+z)) + \xi_{\nu_i}} + 1} + \frac{1}{e^{p/(T_{\nu}(1+z)) - \xi_{\nu_i}} + 1} \right) dp, \quad (\text{A9})$$

$$\pi^2 \xi_{\nu_1} + \xi_{\nu_1}^3 = - \left(\frac{(c_{23}^2 - s_{23}^2)(1 - t_{13}^2 + t_{12}^2 t_{13}^2) + 4t_{12}t_{13}^2 s_{13} s_{23} c_{23}}{(c_{23}^2 - s_{23}^2)(1 - t_{12}^2) - 4s_{13} s_{23} c_{23} t_{12}} + \frac{s_{13}^2}{c_{13}^2 c_{12}^2} \right) (\pi^2 \xi_{\nu_3} + \xi_{\nu_3}^3) + \frac{\pi^2 \xi_{\nu_e} + \xi_{\nu_e}^3}{c_{13}^2 c_{12}^2}, \quad (\text{A10})$$

$$\pi^2 \frac{d\xi_{\nu_1}}{d\xi_{\nu_3}} + 3\xi_{\nu_1}^2 \frac{d\xi_{\nu_1}}{d\xi_{\nu_3}} = - \left(\frac{(c_{23}^2 - s_{23}^2)(1 - t_{13}^2 + t_{12}^2 t_{13}^2) + 4t_{12}t_{13}^2 s_{13} s_{23} c_{23}}{(c_{23}^2 - s_{23}^2)(1 - t_{12}^2) - 4s_{13} s_{23} c_{23} t_{12}} + \frac{s_{13}^2}{c_{13}^2 c_{12}^2} \right) (\pi^2 + 3\xi_{\nu_3}^2), \quad (\text{A11})$$

$$\frac{d\xi_{\nu_1}}{d\xi_{\nu_3}} = \frac{- \left(\frac{(c_{23}^2 - s_{23}^2)(1 - t_{13}^2 + t_{12}^2 t_{13}^2) + 4t_{12}t_{13}^2 s_{13} s_{23} c_{23}}{(c_{23}^2 - s_{23}^2)(1 - t_{12}^2) - 4s_{13} s_{23} c_{23} t_{12}} + \frac{s_{13}^2}{c_{13}^2 c_{12}^2} \right) (\pi^2 + 3\xi_{\nu_3}^2)}{\pi^2 + 3\xi_{\nu_1}^2}, \quad (\text{A12})$$

$$\frac{d^2 \xi_{\nu_1}}{d\xi_{\nu_3}^2} = - \left(\frac{(c_{23}^2 - s_{23}^2)(1 - t_{13}^2 + t_{12}^2 t_{13}^2) + 4t_{12}t_{13}^2 s_{13} s_{23} c_{23}}{(c_{23}^2 - s_{23}^2)(1 - t_{12}^2) - 4s_{13} s_{23} c_{23} t_{12}} + \frac{s_{13}^2}{c_{13}^2 c_{12}^2} \right) \frac{6\xi_{\nu_3}(\pi^2 + 3\xi_{\nu_1}^2) - (\pi^2 + 3\xi_{\nu_3}^2) 6\xi_{\nu_1} \frac{d\xi_{\nu_1}}{d\xi_{\nu_3}}}{(\pi^2 + 3\xi_{\nu_1}^2)^2}, \quad (\text{A13})$$

$$\pi^2 \xi_{\nu_2} + \xi_{\nu_2}^3 = \frac{(c_{23}^2 - s_{23}^2)(1 - t_{13}^2 + t_{12}^2 t_{13}^2) + 4t_{12}t_{13}^2 s_{13} s_{23} c_{23}}{(c_{23}^2 - s_{23}^2)(1 - t_{12}^2) - 4s_{13} s_{23} c_{23} t_{12}} (\pi^2 \xi_{\nu_3} + \xi_{\nu_3}^3), \quad (\text{A14})$$

$$\frac{d\xi_{\nu_2}}{d\xi_{\nu_3}} = \frac{\frac{(c_{23}^2 - s_{23}^2)(1 - t_{13}^2 + t_{12}^2 t_{13}^2) + 4t_{12}t_{13}^2 s_{13} s_{23} c_{23}}{(c_{23}^2 - s_{23}^2)(1 - t_{12}^2) - 4s_{13} s_{23} c_{23} t_{12}} (\pi^2 + 3\xi_{\nu_3}^2)}{\pi^2 + 3\xi_{\nu_2}^2}, \quad (\text{A15})$$

$$\frac{d^2 \xi_{\nu_2}}{d\xi_{\nu_3}^2} = \frac{(c_{23}^2 - s_{23}^2)(1 - t_{13}^2 + t_{12}^2 t_{13}^2) + 4t_{12}t_{13}^2 s_{13} s_{23} c_{23}}{(c_{23}^2 - s_{23}^2)(1 - t_{12}^2) - 4s_{13} s_{23} c_{23} t_{12}} \frac{6\xi_{\nu_3}(\pi^2 + 3\xi_{\nu_2}^2) - (\pi^2 + 3\xi_{\nu_3}^2) 6\xi_{\nu_2} \frac{d\xi_{\nu_2}}{d\xi_{\nu_3}}}{(\pi^2 + 3\xi_{\nu_2}^2)^2}, \quad (\text{A16})$$

$$(\text{A17})$$

4. Angle Perturbation

$$\delta\beta_{\text{avg}}(z) = 0 \quad (\text{A18})$$

$$0 = \frac{z}{3}\beta_{\text{avg}}^{-2}(z) \left(2\beta_{\perp}(z)\delta\beta_{\perp}(z)\beta_{\parallel}(z) + \beta_{\perp}^2(z)\delta\beta_{\parallel}(z) \right), \quad (\text{A19})$$

$$\delta\beta_{\perp}(z) = -\frac{\delta r_d}{r_d^2}\chi(z) + \frac{1}{r_d}\delta\chi(z), \quad (\text{A20})$$

$$\delta\beta_{\parallel}(z) = -\frac{\delta r_d}{r_d^2 H(z)} - \frac{\delta H(z)}{r_d(H(z))^2}, \quad (\text{A21})$$

$$\delta\frac{1}{\theta_{MC}} = -\frac{\delta r_*}{r_*^2}\chi(z_*) + \frac{1}{r_*}\delta\chi(z_*), \quad (\text{A22})$$

$$\delta\chi(z) = \int_0^z dz' \left(-\frac{1}{H^2(z')} \right) \delta H(z'), \quad (\text{A23})$$

$$\delta r_d = \int_{z_d}^{\infty} dz' \left(-\frac{c_s(z')}{H^2(z')} \right) \delta H(z'), \quad (\text{A24})$$

$$\delta r_* = \int_{z_*}^{\infty} dz' \left(-\frac{c_s(z')}{H^2(z')} \right) \delta H(z'), \quad (\text{A25})$$

$$\delta H(z) = \frac{H_0^2}{2H(z)} \left(\frac{\delta\rho_r(z)}{\rho_{\text{crit}}} + \Omega_{\Lambda} e^{3(1+w_0+w_a)\ln(1+z)+3w_a(\frac{1}{1+z}-1)} \left[3\ln(1+z)\delta w_0 + 3(\ln(1+z) + \frac{1}{1+z} - 1)\delta w_a \right] \right), \quad (\text{A26})$$

$$\delta\rho_r(z) = \sum_i \frac{1}{2\pi^2} \int_0^{\infty} \sqrt{p^2 + \left(\frac{M_{\nu}}{3}\right)^2} p^2 \left(\frac{-e^{p/(T_{\nu}(1+z))+\xi_{\nu_i}}}{(e^{p/(T_{\nu}(1+z))+\xi_{\nu_i}} + 1)^2} \delta\xi_{\nu_i} + \frac{e^{p/(T_{\nu}(1+z))-\xi_{\nu_i}}}{(e^{p/(T_{\nu}(1+z))-\xi_{\nu_i}} + 1)^2} \delta\xi_{\nu_i} \right) \quad (\text{A27})$$

$$+ \frac{M_{\nu}\delta M_{\nu}}{9\sqrt{p^2 + \left(\frac{M_{\nu}}{3}\right)^2}} p^2 \left(\frac{1}{e^{p/(T_{\nu}(1+z))+\xi_{\nu_i}} + 1} + \frac{1}{e^{p/(T_{\nu}(1+z))-\xi_{\nu_i}} + 1} \right) dp, \quad (\text{A28})$$

$$(\text{A29})$$

5. Comparing the Coefficients with BAO Angle Perturbation

$$0 = A_d \delta w_0 + B_d \delta w_a + C_d \frac{\delta M_\nu}{M_\nu^{ref}} + D_d \delta \xi_{\nu_3} + E_d \delta \xi_{\nu_3}^2, \quad (\text{A30})$$

$$A_d = \frac{z\beta_\perp(z)}{3\beta_{\text{avg}}^2(z)} \left(2\beta_\parallel(z) \left(\frac{\chi(z)}{r_d^2} \int_{z_d}^\infty dz' c_s(z') K_A(z') - \frac{1}{r_d} \int_0^z dz' K_A(z') \right) + \beta_\perp(z) \left(\frac{1}{r_d^2 H(z)} \int_{z_d}^\infty dz' c_s(z') K_A(z') - \frac{K_A(z)}{r_d} \right) \right), \quad (\text{A31})$$

$$B_d = \frac{z\beta_\perp(z)}{3\beta_{\text{avg}}^2(z)} \left(2\beta_\parallel(z) \left(\frac{\chi(z)}{r_d^2} \int_{z_d}^\infty dz' c_s(z') K_B(z') - \frac{1}{r_d} \int_0^z dz' K_B(z') \right) + \beta_\perp(z) \left(\frac{1}{r_d^2 H(z)} \int_{z_d}^\infty dz' c_s(z') K_B(z') - \frac{K_B(z)}{r_d} \right) \right), \quad (\text{A32})$$

$$C_d = \frac{z\beta_\perp(z) M_\nu^{ref}}{3\beta_{\text{avg}}^2(z)} \left(2\beta_\parallel(z) \left(\frac{\chi(z)}{r_d^2} \int_{z_d}^\infty dz' c_s(z') K_C(z') - \frac{1}{r_d} \int_0^z dz' K_C(z') \right) + \beta_\perp(z) \left(\frac{1}{r_d^2 H(z)} \int_{z_d}^\infty dz' c_s(z') K_C(z') - \frac{K_C(z)}{r_d} \right) \right), \quad (\text{A33})$$

$$D_d = \frac{z\beta_\perp(z)}{3\beta_{\text{avg}}^2(z)} \left(2\beta_\parallel(z) \left(\frac{\chi(z)}{r_d^2} \int_{z_d}^\infty dz' c_s(z') K_D(z') - \frac{1}{r_d} \int_0^z dz' K_D(z') \right) + \beta_\perp(z) \left(\frac{1}{r_d^2 H(z)} \int_{z_d}^\infty dz' c_s(z') K_D(z') - \frac{K_D(z)}{r_d} \right) \right), \quad (\text{A34})$$

$$E_d = \frac{1}{2} \frac{\partial D_d}{\partial \xi_{\nu_3}}, \quad (\text{A35})$$

$$= \frac{1}{2} \left(\frac{\partial D_d}{\partial K_D(z')} \frac{\partial K_D(z')}{\partial \xi_{\nu_3}} + \frac{\partial D_d}{\partial K_D(z)} \frac{\partial K_D(z)}{\partial \xi_{\nu_3}} + \frac{\partial D_d}{\partial \beta_\parallel} \frac{\partial \beta_\parallel}{\partial \xi_{\nu_3}} + \frac{\partial D_d}{\partial \beta_\perp} \frac{\partial \beta_\perp}{\partial \xi_{\nu_3}} + \frac{\partial D_d}{\partial \beta_{\text{avg}}} \frac{\partial \beta_{\text{avg}}}{\partial \xi_{\nu_3}} + \frac{\partial D_d}{\partial \chi} \frac{\partial \chi}{\partial \xi_{\nu_3}} + \frac{\partial D_d}{\partial r_d} \frac{\partial r_d}{\partial \xi_{\nu_3}} + \frac{\partial D_d}{\partial H} \frac{\partial H}{\partial \xi_{\nu_3}} \right), \quad (\text{A36})$$

$$K_A(z) = \frac{H_0^2}{2H^3(z)} 3 \ln(1+z) \Omega_\Lambda e^{3(1+w_0+w_a) \ln(1+z) + 3w_a(\frac{1}{1+z} - 1)}, \quad (\text{A37})$$

$$K_B(z) = \frac{H_0^2}{2H^3(z)} 3 \left(\ln(1+z) + \frac{1}{1+z} - 1 \right) \Omega_\Lambda e^{3(1+w_0+w_a) \ln(1+z) + 3w_a(\frac{1}{1+z} - 1)}, \quad (\text{A38})$$

$$K_C(z) = \frac{H_0^2}{2H^3(z)} \sum_i \frac{1}{2\pi^2} \int_0^\infty \frac{M_\nu}{9\sqrt{p^2 + \left(\frac{M_\nu}{3}\right)^2}} p^2 \left(\frac{1}{e^{p/((1+z)T_\nu) + \xi_{\nu_i}} + 1} + \frac{1}{e^{p/((1+z)T_\nu) - \xi_{\nu_i}} + 1} \right) dp, \quad (\text{A39})$$

$$K_D(z) = \frac{H_0^2}{2H^3(z)} \sum_i \frac{1}{2\pi^2} \int_0^\infty \sqrt{p^2 + \left(\frac{M_\nu}{3}\right)^2} p^2 \frac{d\xi_{\nu_i}}{d\xi_{\nu_3}} \left(\frac{-e^{p/((1+z)T_\nu) + \xi_{\nu_i}}}{(e^{p/((1+z)T_\nu) + \xi_{\nu_i}} + 1)^2} + \frac{e^{p/((1+z)T_\nu) - \xi_{\nu_i}}}{(e^{p/((1+z)T_\nu) - \xi_{\nu_i}} + 1)^2} \right) dp, \quad (\text{A40})$$

$$(\text{A41})$$

The components of E_d are as follows:

$$\frac{\partial D_d}{\partial K_D(z')} = \frac{z\beta_\perp(z)}{3\beta_{\text{avg}}^2(z)} \left(2\beta_\parallel(z) \left(\frac{\chi(z)}{r_d^2} \int_{z_d}^\infty dz' c_s(z') - \frac{1}{r_d} \int_0^z dz' \right) + \beta_\perp(z) \left(\frac{1}{r_d^2 H(z)} \int_{z_d}^\infty dz' c_s(z') \right) \right), \quad (\text{A42})$$

$$\frac{\partial K_D(z')}{\partial \xi_{\nu_3}} = \frac{H_0^2}{2H^3(z')} \frac{1}{2\pi^2} \int_0^\infty dp p^2 \sqrt{p^2 + \left(\frac{M_\nu}{3} \right)^2} \mathcal{G}(p, z') - 3K_D^2(z)H(z'), \quad (\text{A43})$$

$$\frac{\partial D_d}{\partial K_D(z)} = -\frac{z\beta_\perp^2(z)}{3\beta_{\text{avg}}^2(z)r_d}, \quad (\text{A44})$$

$$\frac{\partial K_D(z)}{\partial \xi_{\nu_3}} = \frac{H_0^2}{2H^3(z)} \frac{1}{2\pi^2} \int_0^\infty dp p^2 \sqrt{p^2 + \left(\frac{M_\nu}{3} \right)^2} \mathcal{G}(p, z) - 3K_D^2(z)H(z), \quad (\text{A45})$$

$$\frac{\partial D_d}{\partial \beta_\parallel(z)} = \frac{z\beta_\perp(z)}{3\beta_{\text{avg}}^2(z)} \left(\frac{2\chi(z)}{r_d^2} \int_{z_d}^\infty dz' c_s(z') K_D(z') - \frac{2}{r_d} \int_0^z dz' K_D(z') \right), \quad (\text{A46})$$

$$\frac{\partial \beta_\parallel(z)}{\partial \xi_{\nu_3}} = \frac{1}{r_d^2 H(z)} \int_{z_d}^\infty dz' c_s(z') K_D(z') - \frac{K_D(z)}{r_d}, \quad (\text{A47})$$

$$\frac{\partial D_d}{\partial \beta_\perp(z)} = \frac{D_d}{\beta_\perp(z)} + \frac{z\beta_\perp(z)}{3\beta_{\text{avg}}^2(z)} \left(\frac{1}{r_d^2 H(z)} \int_{z_d}^\infty dz' c_s(z') K_D(z') - \frac{K_D(z)}{r_d} \right), \quad (\text{A48})$$

$$\frac{\partial \beta_\perp(z)}{\partial \xi_{\nu_3}} = \frac{\chi(z)}{r_d^2} \int_{z_d}^\infty dz' c_s(z') K_D(z') - \frac{1}{r_d} \int_0^z dz' K_D(z'), \quad (\text{A49})$$

$$\frac{\partial D_d}{\partial \beta_{\text{avg}}(z)} = \frac{-2D_d}{\beta_{\text{avg}}(z)}, \quad (\text{A50})$$

$$\frac{\partial \beta_{\text{avg}}(z)}{\partial \xi_{\nu_3}} = D_d, \quad (\text{A51})$$

$$\frac{\partial D_d}{\partial \chi(z)} = \frac{2z\beta_\perp(z)\beta_\parallel(z)}{3\beta_{\text{avg}}^2(z)r_d^2} \int_{z_d}^\infty dz' c_s(z') K_D(z'), \quad (\text{A52})$$

$$\frac{\partial \chi(z)}{\partial \xi_{\nu_3}} = -\int_0^z dz' K_D(z'), \quad (\text{A53})$$

$$\frac{\partial D_d}{\partial r_d} = \frac{z\beta_\perp(z)}{3\beta_{\text{avg}}^2(z)} \quad (\text{A54})$$

$$\times \left(2\beta_\parallel(z) \left(\frac{-2\chi(z)}{r_d^3} \int_{z_d}^\infty dz' c_s(z') K_D(z') + \frac{1}{r_d^2} \int_0^z dz' K_D(z') \right) + \beta_\perp(z) \left(\left(\frac{-2}{r_d^3 H(z)} \int_{z_d}^\infty dz' c_s(z') K_D(z') \right) + \frac{K_D(z)}{r_d^2} \right) \right) \quad (\text{A55})$$

$$\frac{\partial r_d}{\partial \xi_{\nu_3}} = -\int_{z_d}^\infty dz' c_s(z') K_D(z'), \quad (\text{A56})$$

$$\frac{\partial D_d}{\partial H(z)} = -\frac{z\beta_\perp^2(z)}{3\beta_{\text{avg}}^2(z)r_d^2 H^2(z)} \int_{z_d}^\infty dz' c_s(z') K_D(z'), \quad (\text{A57})$$

$$\frac{\partial H(z)}{\partial \xi_{\nu_3}} = H^2(z) K_D(z), \quad (\text{A58})$$

$$\mathcal{G}(p, z) = \sum_{i=1}^3 \frac{d^2 \xi_{\nu_i}}{d\xi_{\nu_3}^2} \left[\frac{e^{p/(T_{\nu 0}(1+z)) - \xi_{\nu_i}}}{(e^{p/(T_{\nu 0}(1+z)) - \xi_{\nu_i}} + 1)^2} - \frac{e^{p/(T_{\nu 0}(1+z)) + \xi_{\nu_i}}}{(e^{p/(T_{\nu 0}(1+z)) + \xi_{\nu_i}} + 1)^2} \right] \quad (\text{A59})$$

$$+ \sum_{i=1}^3 \left(\frac{d\xi_{\nu_i}}{d\xi_{\nu_3}} \right)^2 \left[\frac{e^{p/(T_{\nu 0}(1+z)) - \xi_{\nu_i}} (e^{p/(T_{\nu 0}(1+z)) - \xi_{\nu_i}} - 1)}{(e^{p/(T_{\nu 0}(1+z)) - \xi_{\nu_i}} + 1)^3} + \frac{e^{p/(T_{\nu 0}(1+z)) + \xi_{\nu_i}} (e^{p/(T_{\nu 0}(1+z)) + \xi_{\nu_i}} - 1)}{(e^{p/(T_{\nu 0}(1+z)) + \xi_{\nu_i}} + 1)^3} \right], \quad (\text{A60})$$

$$(\text{A61})$$

Note that the function of z' implies that it is part of the integrand of the integral $\int dz'$.